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Department of Electronics & Communication Engineering

Helixo AI

AI-Based Helmet Detection & Engine Safety Lock System

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THE PROBLEM

Why This Project Matters

- ▶ India has one of the highest road accident fatality rates globally
- ▶ Over 1.5 lakh deaths annually — ~40% are two-wheeler riders
- ▶ Helmet compliance among two-wheeler riders remains critically low
- ▶ Current system is penalty-based — fines are imposed AFTER violation
- ▶ No real-time prevention mechanism exists to stop unsafe riding
- ▶ Riders are fined ₹1,000 under MV Act, yet violations continue
- ▶ Rural areas have minimal traffic enforcement coverage

The Core Problem

- ▶ Reactive System: Penalty after accident/violation
- ▶ No Prevention: Engine starts regardless of helmet
- ▶ Human Dependency: Relies on police presence
- ▶ Zero Automation: No technology-driven enforcement

Our Approach: Shift from Reactive → Preventive

Integrate AI-based helmet detection directly into the vehicle to prevent ignition without helmet usage.

OBJECTIVES

What We Aim to Achieve

Primary Objective

Develop an AI-based system that ensures helmet usage by preventing motorcycle ignition without a helmet and monitoring rider safety in real-time.

Ignition Control

Implement relay-based ignition control that blocks engine start when helmet is not detected ($\geq 70\%$ confidence threshold over multiple frames).

Emergency Override

Provide time-limited, internally logged emergency bypass for critical situations without compromising overall safety.

Real-Time Detection

Use camera-based detection with YOLO AI model to identify helmet presence with high accuracy using sliding window confidence analysis.

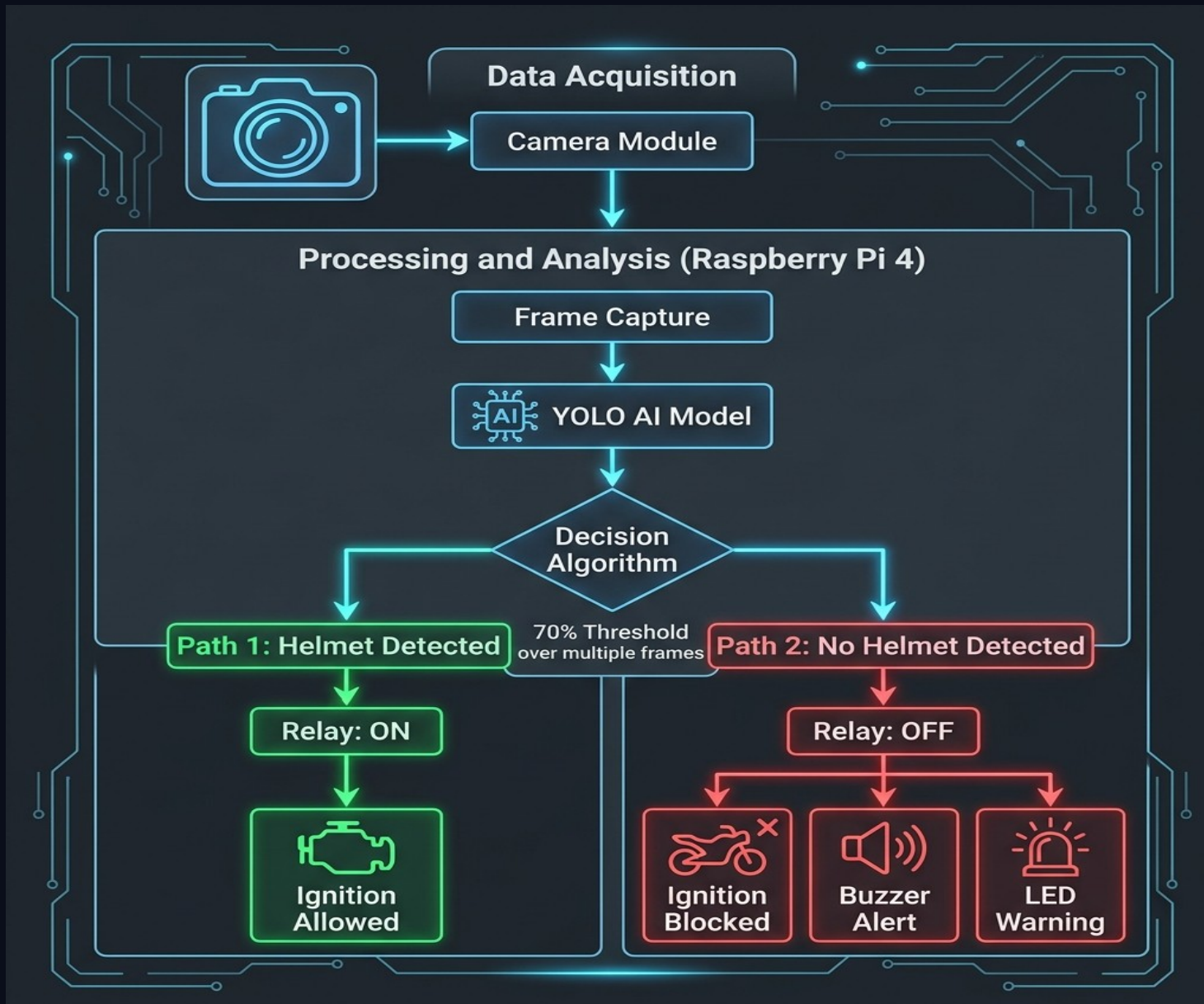
Continuous Monitoring

Monitor helmet usage during the ride with gradual speed reduction and controlled stop if helmet is removed mid-ride.

Cost Optimization

Transition from Raspberry Pi prototype (₹7,000–₹10,500) to embedded ESP32 product (₹1,500–₹2,000 manufacturing cost).

System Architecture & Data Flow



System Flow Overview

- ▶ Camera Module captures rider image at handlebar level
- ▶ Raspberry Pi 4 processes frames using YOLO AI model
- ▶ Each frame classified: Helmet Detected / No Helmet
- ▶ Decision Algorithm uses sliding window (10-sec rolling)
- ▶ Confidence = (Helmet frames / Total frames) × 100
- ▶ If Confidence ≥ 70%: Relay ON → Ignition Allowed ✓
- ▶ If Confidence < 70%: Relay OFF → Ignition Blocked ✗
- ▶ Buzzer + LED provide audio-visual feedback to rider
- ▶ Multi-layer confirmation prevents false positives
 - 3 consecutive NO detections required before action
 - Grace period prevents accidental triggers

AI Detection & Decision Algorithm

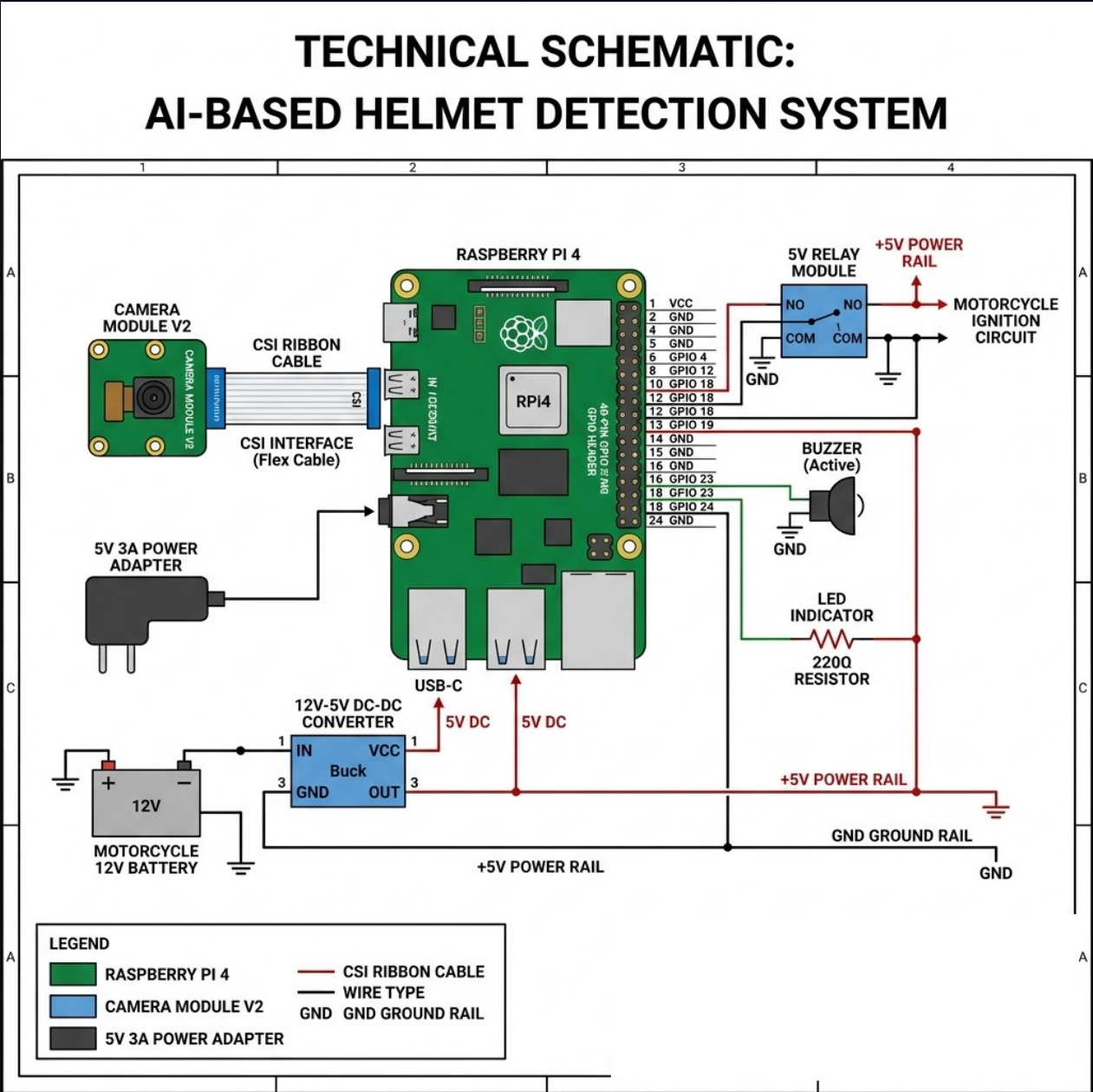
WORKING METHODOLOGY OF AI HELMET DETECTION SYSTEM



Detection Pipeline

- ▶ Camera captures frames at ~15 FPS continuously
- ▶ YOLO (You Only Look Once) AI model processes each frame
 - Pre-trained on helmet/no-helmet dataset
 - Custom training with Indian road conditions
- ▶ Sliding Window Analysis (10-second rolling window):
 - ~150 frames analyzed in each window
 - Each frame gets binary output: Helmet=1 / No=0
- ▶ Confidence Calculation:
 - $\text{Confidence\%} = (\text{Helmet frames} \div \text{Total frames}) \times 100$
- ▶ Threshold-Based Decision:
 - $\geq 70\%$ → SAFE (Green LED) → Ignition Allowed
 - $< 70\%$ → UNSAFE (Red LED) → Ignition Blocked
- ▶ Consistency Check (added safety):
 - Last 3 readings must agree before final action
 - Prevents noise and momentary glitches

Hardware Schematic & Wiring



Complete circuit schematic: Raspberry Pi 4 → Camera V2 (CSI) → 5V Relay (GPIO) → Motorcycle Ignition | 12V Battery → DC Converter → 5V Power Rail

Key Components & Specifications

HARDWARE COMPONENTS FOR AI-BASED HELMET DETECTION SYSTEM

TECH PRESENTATION STYLE



1) RASPBERRY PI 4 MODEL B (4GB RAM) - MAIN PROCESSING UNIT



2) PI CAMERA MODULE V2 (8MP SONY IMX219) - IMAGE CAPTURE



3) 5V RELAY MODULE - IGNITION CONTROL



2) PI CAMERRA MODULE V2 (8MP SONY IMX219) - IMAGE CAPTURE



4) ESP32-S3 ESP32-S3 DEVELOPMENT BOARD - EMBEDDED MCU



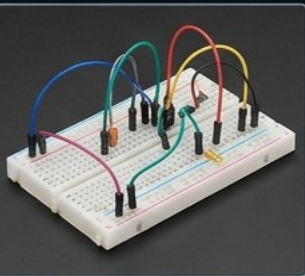
5) OV2640 IR CAMERA - NIGHT VISION CAPABLE



6) BUZZER AND LED INDICATORS - ALERTS



7) 12V TO 5V DC-DC CONVERTER - POWER MANAGEMENT



8) BREADBOARD WITH JUMPER WIRES - PROTOTYPING

Component Specifications

- ▶ **Raspberry Pi 4 (4GB)**
Main processing unit — runs YOLO model, handles camera I/O and GPIO control
- ▶ **Pi Camera Module V2**
8MP Sony IMX219 sensor — captures real-time rider images via CSI interface
- ▶ **5V Relay Module**
Controls motorcycle ignition circuit — switches ON/OFF based on AI decision
- ▶ **ESP32-S3 MCU**
Embedded AI microcontroller for final product phase — runs TensorFlow Lite
- ▶ **OV2640/OV3660 IR Camera**
Night-vision capable camera for all-weather detection
- ▶ **12V-5V DC Converter**
Powers system from motorcycle's 12V battery via buck converter
- ▶ **Buzzer + LED Kit**
Audio-visual indicators — Green (safe), Red (blocked), Buzzer (alert)
- ▶ **Breadboard + Jumper Wires**
Prototyping platform for Phase 1 development and testing

Technology Stack & AI Pipeline



AI & Machine Learning

- ▶ YOLO (You Only Look Once) — Object detection model
 - YOLOv5 for prototype (full model on Raspberry Pi)
 - TensorFlow Lite for embedded ESP32 (optimized)
- ▶ Custom dataset: Indian riders with/without helmets
 - Multiple helmet types: Half, Full-face, Visor up/down
 - Various lighting: Day, Night, Indoor, Outdoor
- ▶ OpenCV for image preprocessing and frame handling
- ▶ Training pipeline: Dataset → Augmentation → Train → Evaluate



Software Architecture

- ▶ Python — Primary language for prototype phase
- ▶ OpenCV — Real-time frame capture and processing
- ▶ GPIO Library — Relay, buzzer, and LED control
- ▶ Decision Engine — Sliding window + threshold logic
- ▶ Logging System — Records override events internally



Decision Algorithm (Pseudocode)

```
INITIALIZE rolling_window = []
INITIALIZE override_active = FALSE

LOOP every 1 second:
    frame = camera.capture()
    result = YOLO_model.detect(frame)

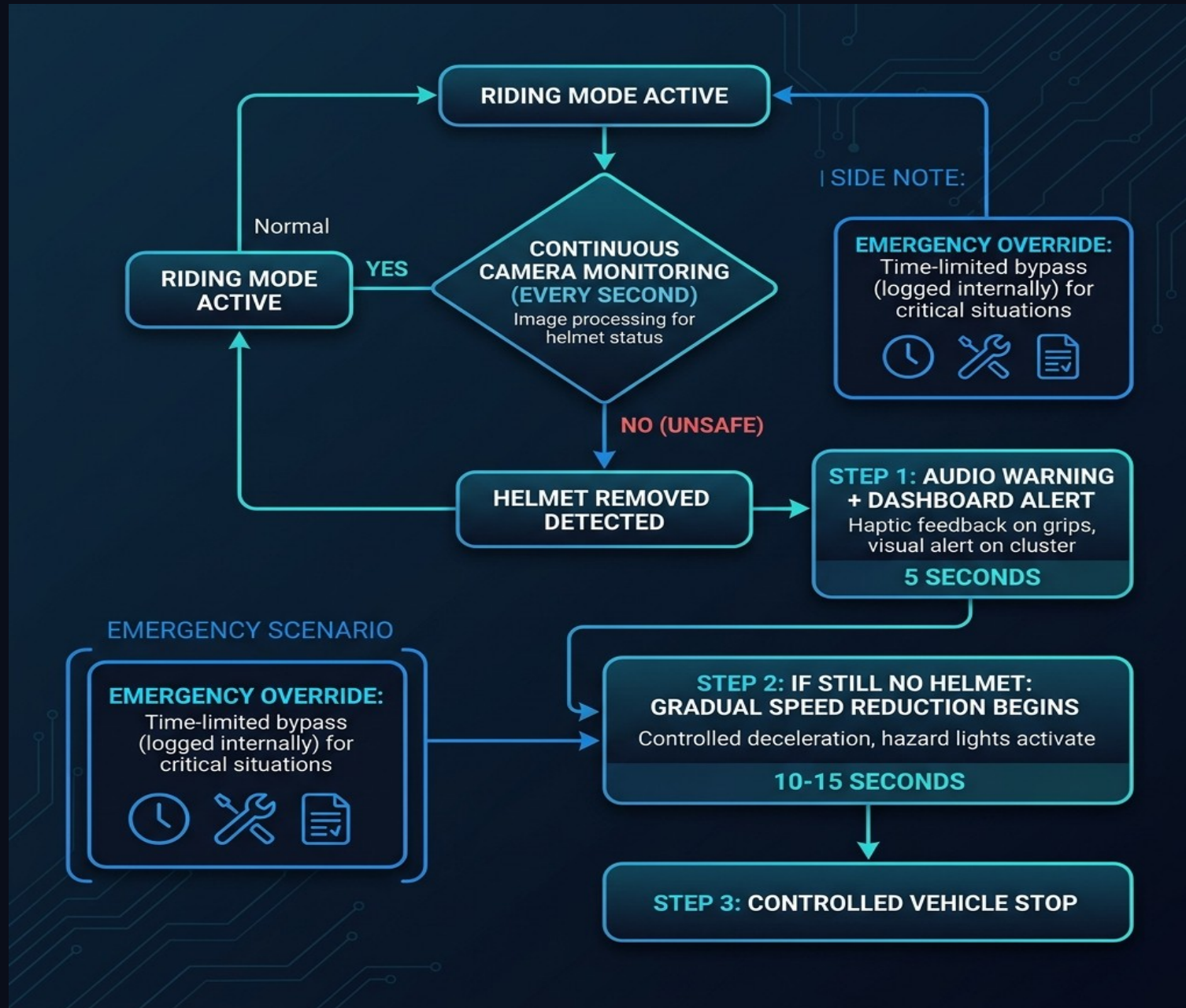
    IF result == HELMET:
        rolling_window.append(1)
    ELSE:
        rolling_window.append(0)

    // Keep last 10 seconds only
    IF len(rolling_window) > 10:
        rolling_window.pop(0)

    confidence = sum(window) / len(window) * 100

    IF confidence >= 70%:
        relay.ON() // Allow ignition
        LED = GREEN
    ELSE:
        IF NOT override_active:
            relay.OFF() // Block ignition
            buzzer.ALERT()
            LED = RED
```

Real-Time Monitoring & Safe Response



Progressive Safety Protocol

► Phase 1: Pre-Start Check (5–10 seconds)

- Camera scans rider before ignition
- $\geq 70\%$ helmet confidence $\rightarrow$ Engine starts
- $< 70\%$ $\rightarrow$ Engine blocked + buzzer alert

► Phase 2: Continuous Monitoring (during ride)

- Every second: update rolling window
- If helmet removed $\rightarrow$ Progressive response:

⚡ Step 1: Warning

Audio buzzer + dashboard LED alert (5 sec grace)

⚡ Step 2: Speed Reduction

Gradual engine power decrease (10-15 sec)

⚡ Step 3: Controlled Stop

Safe vehicle stop — NOT sudden engine cut

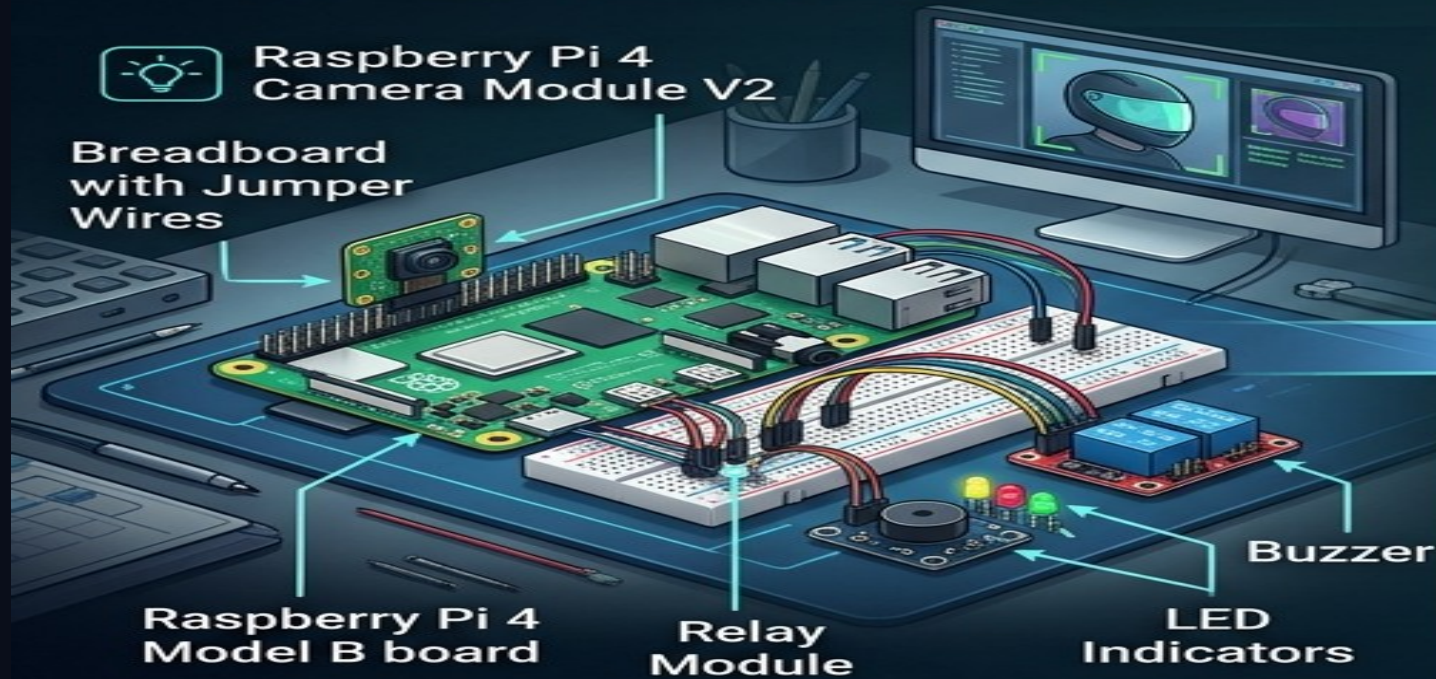
🚨 Emergency Override Feature

Time-limited bypass (logged internally) for critical situations. Prevents misuse while ensuring flexibility. Override events stored with timestamp for audit.

Prototype to Production Architecture

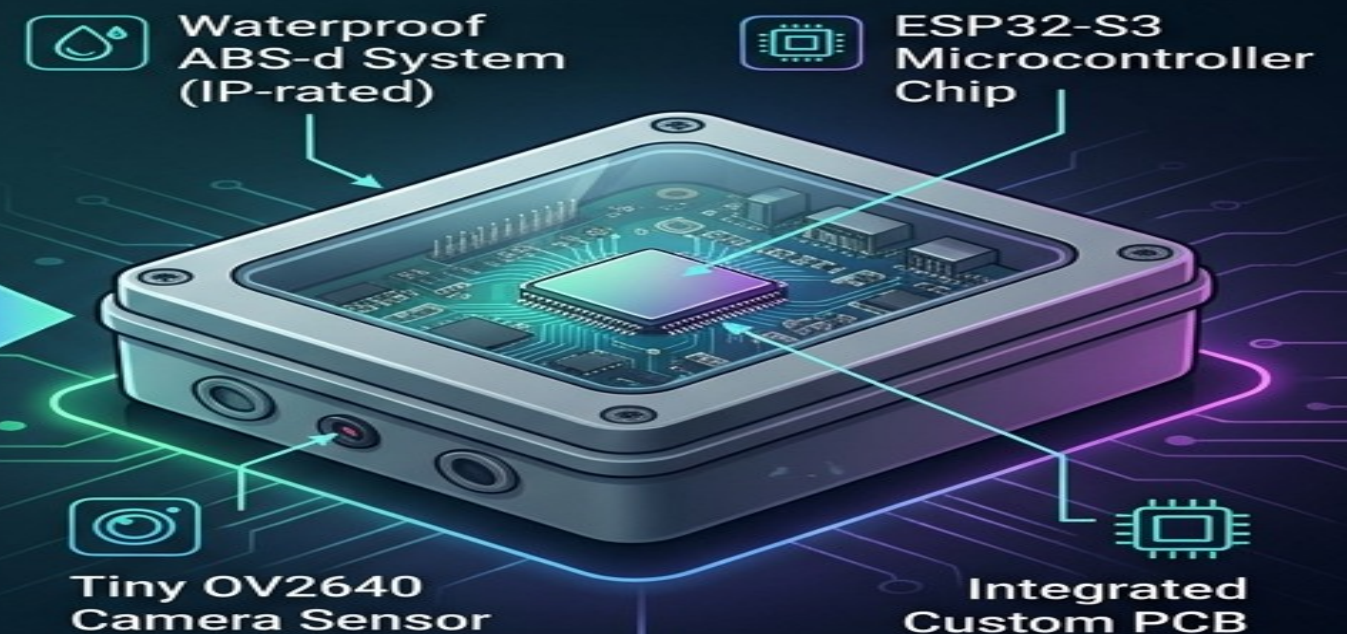
EVOLUTION OF AI HELMET DETECTION SYSTEM: FROM PROTOTYPE TO FINAL PRODUCT

PHASE 1: PROTOTYPE DEVELOPMENT & TESTING



Raspberry Pi 4 Based, YOLOv5 Full Model, Software Processing, Cost: ₹7,000-₹10,500

PHASE 2: FINAL PRODUCT COMMERCIAL DEPLOYMENT



ESP32 Embedded, TensorFlow Lite Optimized, Firmware-based, Manufacturing Cost: ₹1,500-₹2,000

Core intelligence remains identical: Camera → AI Detection → Decision → Relay Control. Only the hardware platform changes between phases.

Raspberry Pi Based Development System

Prototype Specifications

- ▶ Processing Unit: Raspberry Pi 4 Model B (4GB RAM)
 - Quad-core Cortex-A72 @ 1.5GHz
 - Runs full YOLOv5 model in Python
- ▶ Camera: Pi Camera V2 (8MP Sony IMX219)
 - CSI interface — low latency, high bandwidth
 - 15 FPS real-time capture capability
- ▶ Storage: 32GB High-Speed MicroSD Card
 - Stores OS, AI model, dataset, and logs
- ▶ Power: 5V 3A Official Adapter (USB-C)
- ▶ Control: 1-Channel 5V Relay Module
 - Switches motorcycle ignition ON/OFF
- ▶ Indicators: Buzzer + LED Kit
 - Green LED = Helmet detected (safe)
 - Red LED = No helmet (blocked)
 - Buzzer = Audio alert on violation
- ▶ Prototyping: 830-point Breadboard + Jumper Wires
- ▶ Enclosure: Acrylic Protective Case

Development Goals

- ▶ ✓ Validate AI helmet detection accuracy
- ▶ ✓ Test real-time ignition control logic
- ▶ ✓ Evaluate sliding window decision system
- ▶ ✓ Train YOLO model on Indian road dataset
- ▶ ✓ Prototype cost: ₹7,000 – ₹10,500

AI Model Details

- ▶ Model: YOLOv5 (full version)
- ▶ Framework: PyTorch → ONNX → Inference
- ▶ Dataset: Custom (helmet/no-helmet images)
 - Indian road conditions, various helmets
 - Day, night, indoor, outdoor lighting
 - Half helmet, full face, visor up/down
- ▶ Expected Accuracy: 85–90%
- ▶ Processing: Software-based (Python + OpenCV)
- ▶ Decision: 70% threshold + consistency check

ESP32 Embedded System for Mass Deployment

Embedded System Specifications

- ▶ **Processing Unit:** ESP32-S3 / STM32 AI MCU
 - ▶ Dual-core Xtensa LX7 @ 240MHz
 - ▶ Built-in AI acceleration capability
 - ▶ Ultra-low power consumption
- ▶ **Camera:** OV2640 / OV3660 IR Camera
 - ▶ Night-vision capable (IR illumination)
 - ▶ Compact form factor for vehicle mounting
- ▶ **PCB:** Custom 4-Layer PCB (production grade)
 - ▶ Integrated all components on single board
 - ▶ SMD assembly for reliability
- ▶ **Power:** 12V to 5V DC Converter (from bike battery)
- ▶ **Control:** Automotive-grade Ignition Relay
 - ▶ Higher current rating for real ignition systems
- ▶ **Enclosure:** Waterproof ABS Casing (IP-rated)
 - ▶ Weather-resistant for outdoor mounting
- ▶ **AI Model:** TensorFlow Lite (optimized/quantized)
 - ▶ Firmware-based — runs directly on MCU
 - ▶ No OS required — instant boot

Key Advantages

- ▶ **Manufacturing Cost:** ₹1,500 – ₹2,000 per unit
- ▶ **Market Price:** ₹2,500 – ₹4,000 (profitable margin)
- ▶ **Compact & Vehicle-Mountable design**
- ▶ **Instant boot** — no OS loading time
- ▶ **IP-rated waterproof enclosure**
- ▶ **Night vision with IR camera support**

Comparison: Prototype vs Product

Feature	Phase 1 (Prototype)	Phase 2 (Product)
Platform	Raspberry Pi 4	ESP32-S3 MCU
Camera	Pi Camera V2	OV2640/IR Camera
AI Model	YOLOv5 (full)	TF Lite (optimized)
Processing	Software (Python)	Firmware (embedded)
Cost	₹7,000–₹10,500	₹1,500–₹2,000
Deployment	Lab/Testing	Mass production
Night Vision	No	Yes (IR camera)
Boot Time	30-60 seconds	Instant

Expected Output & Performance Metrics

Expected System Outputs

- ✓ **Helmet Detected**
Bike ignition starts normally — Green LED on, no buzzer
- ✗ **No Helmet Detected**
Ignition blocked — Red LED on, buzzer alerts rider
- ⚡ **Helmet Removed Mid-Ride**
Progressive: Warning → Speed reduction → Controlled stop
- 🔓 **Emergency Override Used**
Time-limited bypass activated — event logged with timestamp
- 📊 **Detection Accuracy**
Expected 85–90% accuracy with custom-trained YOLO model
- ⚡ **Response Time**
< 1 second detection latency with 15 FPS camera feed

Project Output Statement

The project will result in a working prototype of an AI-based helmet enforcement system capable of real-time detection and control. It demonstrates prevention-based safety rather than penalty-based enforcement. The system showcases scalability by transitioning from a prototype platform to an embedded system, reducing cost significantly (₹1,000–₹2,000 per vehicle).

Advantages, Limitations & Solutions

✓ Advantages

- ▶ Preventive Safety: Stops accidents BEFORE they happen
- ▶ No Helmet Dependency: Works with any helmet type
 - No RFID tags, Bluetooth modules, or pairing needed
- ▶ Real-Time AI: Continuous monitoring, not one-time check
- ▶ Low Cost: Final product at ₹1,500–₹2,000 per unit
- ▶ Easy Installation: Retrofit kit for existing bikes
- ▶ Scalable: OEM integration possible for new vehicles
- ▶ Fail-Safe Design: Gradual response, not sudden cutoff
- ▶ Tamper-Resistant: Camera-based, no easy workaround
- ▶ Future-Ready: Can integrate GPS, cloud, mobile app

⚠ Limitations & Solutions

- ▶ **Night Detection Challenge**
Without proper lighting, accuracy drops
→ **Solution: IR/Night-vision camera (OV2640 IR)**
- ▶ **Weather Dependency**
Rain, dust, fog can affect camera
→ **Solution: Waterproof casing + camera cleaning mechanism**
- ▶ **Helmet Type Confusion**
Half helmet vs cap vs turban
→ **Solution: Extensive dataset training with Indian headwear**
- ▶ **Model Accuracy**
No AI is 100% — false positives possible
→ **Solution: Multi-layer confirmation (3 consecutive checks)**
- ▶ **Processing Power**
Raspberry Pi has limited edge AI capability
→ **Solution: ESP32-S3 with TensorFlow Lite optimization**
- ▶ **Power Consumption**
Continuous AI processing draws power
→ **Solution: Low-power modes + efficient model quantization**

Development Timeline & Methodology

Phase	Activity	Details
Phase 1	Problem Study & Requirement Analysis	Traffic safety study, Helmet violation analysis, Existing systems review, System requirements finalization, Feasibility study
Phase 2	System Design & Architecture	Block diagram design, AI model flow design, Hardware-software integration plan, Sensor/camera placement, Final system blueprint
Phase 3	Dataset Preparation & Model Training	Dataset collection (helmet/no-helmet), Data cleaning & labeling, Model selection (YOLO/CNN), Training AI model, Accuracy testing & tuning
Phase 4	System Integration & Development	AI model integration, Real-time detection setup, Vehicle control logic, Alert & ignition control, Debugging integration errors
Phase 5	Testing, Optimization & Deployment	Real-world testing, Error detection & fixing, Performance optimization, Safety validation tests, Final deployment & documentation

Roadmap for Enhancement & Expansion

GPS Integration

Location tracking and geo-fenced enforcement zones. Track violation patterns by area for government analytics.

Insurance Discounts

Compliance data shared with insurance companies for premium reduction benefits to responsible riders.

Smart City Integration

Feed safety data to urban traffic management systems and municipal safety dashboards for city-wide analytics.

Driver Monitoring

Expand AI capabilities to detect fatigue, distraction, drowsiness, and other safety risks beyond helmet detection.

Cloud Connectivity

Upload violation logs, compliance reports, and system health data to cloud servers for fleet management.

Mobile App Integration

Companion app for ride history, safety score, compliance reports, and real-time alerts to vehicle owner.

OEM Integration

Partner with two-wheeler manufacturers (Hero, TVS, Bajaj) for factory-level integration in new vehicles.

Advanced AI Models

Upgrade to more sophisticated models: face recognition, multiple rider detection, helmet type classification.

Summary & Impact

► Real-Time AI Enforcement

Prevents ignition without a helmet — shifting enforcement from reactive (penalty-based) to preventive (technology-based). No human intervention required.

► Progressive Response System

Gradual speed reduction and controlled stop mechanism ensures safety without creating dangerous situations from sudden engine cutoffs.

► Scalable Two-Phase Architecture

Prototype validates the concept on Raspberry Pi; embedded ESP32 final product enables mass deployment at ₹1,500–₹2,000 manufacturing cost.

► Intelligent Safety Design

Multi-layer confidence system with 70% threshold, sliding window analysis, and consistency checks ensures reliable detection with minimal false triggers.

► Emergency Override Usability

Time-limited, internally logged bypass ensures real-world usability and flexibility without compromising overall system safety.

► Revolution in Two-Wheeler Safety

This project has the potential to become a future mandatory safety standard in motorcycles, similar to the evolution of seatbelt alerts, airbags, and ABS systems.

"This project represents a shift from reactive safety systems to proactive safety enforcement — a potential revolution in two-wheeler safety."

Component Budget & Grant Requirement

Sr. No.	Item / Component	Qty	Cost/Item (₹)	Total Cost (₹)
1	Raspberry Pi 4 Model B (4GB)	1	6,500	6,500
2	Raspberry Pi Camera V2 (8MP Sony IMX219)	1	2,300	2,300
3	32GB MicroSD Card (High Speed)	1	800	800
4	5V 3A Official Adapter	1	800	800
5	1-Channel 5V Relay	1	200	200
6	Indicator Kit	1	150	150
7	830-point Breadboard	1	300	300
8	Mixed Set (Jumper Wires)	1	300	300
9	Camera Ribbon Cable	1	250	250
10	Acrylic Protective Case	1	500	500
11	ESP32-S3 / STM32 AI MCU	1	900	900
12	OV2640 / OV3660 IR Camera	1	800	800
13	Custom 4-Layer PCB (Prototype batch)	1	500	500
14	12V to 5V DC Converter	1	200	200
15	Automotive Ignition Relay	1	250	250
16	Automotive Grade Set	1	200	200
17	Industrial Indicator Set	1	100	100
18	Waterproof ABS Casing	1	600	600
TOTAL				₹15,650

We request a grant of ₹15,650 (approx) for the complete development of this AI-based Helmet Detection & Engine Safety Lock System prototype and final embedded product.

Thank You

Questions & Answers

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